

Cal 2 - Exam Question 34

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Question 34

Evaluate

$$\int \frac{x^2}{(4-x^2)^{\frac{3}{2}}} dx.$$

- a. $\frac{2x}{\sqrt{4-x^2}} - 2 \sin^{-1}\left(\frac{x}{2}\right) + C$
- b. $\frac{x}{\sqrt{4-x^2}} - \sin^{-1}\left(\frac{x}{2}\right) + C$
- c. $\frac{4}{\sqrt{4-x^2}} - 2 \sin^{-1}\left(\frac{x}{2}\right) + C$
- d. I do not know

Solution: b) $\frac{x}{\sqrt{4-x^2}} - \sin^{-1}\left(\frac{x}{2}\right) + C$

This question is asking you to: Compute an [antiderivative](#) by rewriting the [denominator](#) into a form that matches a standard [trigonometric substitution](#) pattern, then substitute back to write the final answer in terms of x .

Step 1: Rewrite the denominator $(4-x^2)^{\frac{3}{2}}$.

Step 1.1:

Identify the [denominator](#)

In

$$\frac{x^2}{(4-x^2)^{\frac{3}{2}}},$$

the denominator is the bottom part:

$$(4-x^2)^{\frac{3}{2}}.$$

Step 1.2:Rewrite [fractional exponents](#)

Since the denominator is complicated, we want to simplify it by rewriting the power.

$$(4 - x^2)^{\frac{3}{2}}$$

has a fractional exponent, and we can rewrite fractional exponents using:

$$a^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m, \quad \sqrt[n]{a} = a^{\frac{1}{n}}.$$

Step 1.3:Rewrite the power $\frac{3}{2}$

$$\begin{aligned} \text{Rewrite } \frac{3}{2} &= \frac{1}{2} \cdot 3 & (4 - x^2)^{\frac{3}{2}} &= (4 - x^2)^{\frac{1}{2} \cdot 3} \\ \text{Use } a^{bc} &= (a^b)^c & (4 - x^2)^{\frac{1}{2} \cdot 3} &= \left((4 - x^2)^{\frac{1}{2}} \right)^3 \\ \text{Rewrite } (4 - x^2)^{\frac{1}{2}} &\text{ as a } \text{square root} & \left((4 - x^2)^{\frac{1}{2}} \right)^3 &= \left(\sqrt{4 - x^2} \right)^3 \end{aligned}$$

Step 2: Set up [trigonometric substitution](#).**Step 2.1:**

Notice the trig substitution pattern

Right now the denominator still looks complicated, so we want to simplify it more. The denominator contains

$$\sqrt{4 - x^2},$$

and this looks like the [trigonometric substitution](#) pattern

$$\sqrt{a^2 - x^2}.$$

Step 2.2:

Define the symbols

a : [constant](#), x : [variable](#), θ : angle.

Step 2.3:

Trigonometric pattern

$\sqrt{a^2 - x^2}$

When we encounter the pattern

$\sqrt{a^2 - x^2},$

we let

$x = a \sin \theta.$

Step 2.4:Rewrite the fraction $\sqrt{a^2 - x^2}$ Substitute $x = a \sin \theta$

$$\sqrt{a^2 - \boxed{x}^2} = \sqrt{a^2 - (\boxed{a \sin \theta})^2}$$

Step 2.5:Simplify $\sqrt{a^2 - (a \sin \theta)^2}$ Use [Power of a Product Rule](#), $(\boxed{a} \boxed{b})^2 = a^2 \boxed{b}^2$ Substitute $b = \sin \theta$, $(\boxed{a \sin \theta})^2 = a^2 (\boxed{\sin \theta})^2$

$$\sqrt{a^2 - \boxed{(a \sin \theta)}^2} = \sqrt{a^2 - \boxed{a^2} (\boxed{\sin \theta})^2}$$

Step 2.6:

Trig substitution summary

For pattern $\sqrt{a^2 - x^2}$, let $x = a \sin \theta$

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 (\sin \theta)^2}$$

Step 3: Rewrite $\sqrt{4 - x^2}$ using trigonometric substitution.**Step 3.1:**Identify a and x in $\sqrt{4 - x^2}$ Rewrite $4 = 2 \cdot 2 = 2^2$

$$\sqrt{\boxed{4} - x^2} = \sqrt{\boxed{2^2} - x^2}$$

$$\text{Let } \sqrt{2^2 - x^2} = \sqrt{a^2 - x^2}$$

$$a = \boxed{2}$$

$$x = \boxed{a} \sin \theta = \boxed{2} \sin \theta$$

Step 3.2:Substitute $a = 2$, $x = 2 \sin \theta$

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2(\sin \theta)^2}$$

into the fraction

$$\sqrt{2^2 - x^2} = \sqrt{2^2 - 2^2(\sin \theta)^2}$$

Step 4: Simplify $\sqrt{2^2 - 2^2(\sin \theta)^2}$.**Step 4.1:**Replace $2^2 = 4$

$$\text{Replace } 2^2 = 4 \quad \sqrt{2^2 - 2^2(\sin \theta)^2} = \sqrt{4 - 4(\sin \theta)^2}$$

Step 4.2:

Factor out 4

$$\text{Rewrite } 4 = 4 \cdot 1 \quad 4 - 4(\sin \theta)^2 = 4(1) - 4((\sin \theta)^2)$$

$$\text{Factor out 4} \quad 4 - 4(\sin \theta)^2 = 4(1 - (\sin \theta)^2)$$

Step 4.3:Simplify $1 - (\sin \theta)^2$

$$\text{By } (\sin \theta)^2 = \sin^2 \theta, \quad 4(1 - (\sin \theta)^2) = 4(1 - \sin^2 \theta)$$

$$\text{By [Pythagorean identity](#),} \quad 1 - \sin^2 \theta = \cos^2 \theta$$

$$\text{Substitute the Pythagorean identity,} \quad 4(1 - \sin^2 \theta) = 4(\cos^2 \theta)$$

Step 4.4:

Summary

$$\sqrt{4 - x^2} = \sqrt{2^2 - x^2} \quad \text{from Step 3.1}$$

$$= \sqrt{2^2 - 2^2(\sin \theta)^2} \quad \text{from Step 3.2}$$

$$= \sqrt{4(1 - \sin^2 \theta)} \quad \text{from Step 4.1 and Step 4.2}$$

$$= \sqrt{4 \cos^2 \theta} \quad \text{from Step 4.3}$$

Step 5: Compute $(4 - x^2)^{\frac{3}{2}}$ **after substitution.**

Step 5.1:Substitute into $(\sqrt{4-x^2})^3$

From Step 1.3, $(4-x^2)^{\frac{3}{2}} = (\sqrt{4-x^2})^3$

Substitute $\sqrt{4-x^2} = \sqrt{4\cos^2\theta}$ $\left(\sqrt{4-x^2}\right)^3 = \left(\sqrt{4\cos^2\theta}\right)^3$

Step 5.2:Simplify $\sqrt{4\cos^2\theta}$

Since $\sqrt{ab} = \sqrt{a}\sqrt{b}$, $\sqrt{4\cos^2\theta} = \sqrt{4}\sqrt{\cos^2\theta}$

$\sqrt{4} = 2$ $\sqrt{4}\sqrt{\cos^2\theta} = 2\sqrt{\cos^2\theta}$

Step 5.3:Use absolute value: $\sqrt{u^2} = |u|$

$u = \cos\theta$

$\sqrt{\cos^2\theta} = |\cos\theta|$

$2\sqrt{\cos^2\theta} = 2|\cos\theta|$

$\sqrt{4\cos^2\theta} = 2|\cos\theta|$

Let $u = \cos\theta$

Use $\sqrt{u^2} = |u|$

Multiply both sides by 2

Step 5.4:

Cube the result

$$\begin{aligned} \left(\sqrt{4\cos^2\theta}\right)^3 &= (2|\cos\theta|)^3 \\ &= 2^3|\cos\theta|^3 \\ &= 8|\cos\theta|^3 \end{aligned}$$

Step 5.5:

Remove the absolute value

$$0 \leq \theta \leq \frac{\pi}{2} \Rightarrow \cos\theta \geq 0$$

$$|\cos\theta| = \cos\theta \text{ since } \cos\theta \geq 0$$

$$8|\cos\theta|^3 = 8\cos^3\theta$$

$$(4-x^2)^{\frac{3}{2}} = 8\cos^3\theta$$

Step 6: Rewrite the numerator x^2 .**Step 6.1:**

Square both sides

$$\begin{aligned} x &= 2\sin\theta \\ x^2 &= (2\sin\theta)^2 \end{aligned}$$

Step 6.2:Apply $(ab)^2 = a^2b^2$

$$a = 2$$

$$b = \sin \theta$$

$$(2 \sin \theta)^2 = 2^2 (\sin \theta)^2$$

Step 6.3:

Simplify

$$2^2 = 4$$

$$(\sin \theta)^2 = \sin^2 \theta \quad (\text{shorthand})$$

$$2^2 (\sin \theta)^2 = 4 \sin^2 \theta$$

Step 7: Compute dx .**Step 7.1:**Differentiate $x = 2 \sin \theta$

$$x = 2 \sin \theta$$

$$\begin{aligned} \frac{dx}{d\theta} &= \frac{d}{d\theta}(2 \sin \theta) \\ &= 2 \frac{d}{d\theta}(\sin \theta) \\ &= 2 \cos \theta \end{aligned}$$

Differentiate both sides with respect to θ
Use constant multiple rule: $\frac{d}{d\theta}(cf(\theta)) = cf'(\theta)$
Plug in $\frac{d}{d\theta}(\sin \theta) = \cos \theta$

Step 7.2:Write dx

$$\frac{dx}{d\theta} = 2 \cos \theta$$

$$d\theta \left(\frac{dx}{d\theta} \right) = (2 \cos \theta) d\theta$$

$$dx = 2 \cos \theta d\theta$$

Step 8: Substitute into the integral.**Step 8.1:**

List substitutions

$$x^2 = 4 \sin^2 \theta$$

$$(4 - x^2)^{\frac{3}{2}} = 8 \cos^3 \theta$$

$$dx = 2 \cos \theta d\theta$$

Step 8.2:

Substitute

$$\int \frac{x^2}{(4 - x^2)^{\frac{3}{2}}} dx = \int \frac{4 \sin^2 \theta}{8 \cos^3 \theta} (2 \cos \theta d\theta)$$

Step 9: Simplify the integrand.

Step 9.1:

Multiply constants

$$\begin{aligned}\int \frac{4 \sin^2 \theta}{8 \cos^3 \theta} 2 \cos \theta \, d\theta &= \int \frac{4 \cdot 2 \sin^2 \theta \cos \theta}{8 \cos^3 \theta} \, d\theta \\ &= \int \frac{8 \sin^2 \theta \cos \theta}{8 \cos^3 \theta} \, d\theta\end{aligned}$$

Step 9.2:

Cancel 8

$$\int \frac{8 \sin^2 \theta \cos \theta}{8 \cos^3 \theta} \, d\theta = \int \frac{\sin^2 \theta \cos \theta}{\cos^3 \theta} \, d\theta$$

Step 9.3:Cancel $\cos \theta$

$$\begin{aligned}\cos^3 \theta &= \cos^2 \theta \cos \theta \\ \int \frac{\sin^2 \theta \cos \theta}{\cos^3 \theta} \, d\theta &= \int \frac{\sin^2 \theta \cos \theta}{\cos^2 \theta \cos \theta} \, d\theta \\ &= \int \frac{\sin^2 \theta}{\cos^2 \theta} \, d\theta\end{aligned}$$

Step 10: Rewrite using tangent.**Step 10.1:**

Use the definition of tangent

$$\begin{aligned}\tan \theta &= \frac{\sin \theta}{\cos \theta} \\ \tan^2 \theta &= \frac{\sin^2 \theta}{\cos^2 \theta} \\ \int \frac{\sin^2 \theta}{\cos^2 \theta} \, d\theta &= \int \tan^2 \theta \, d\theta\end{aligned}$$

Step 11: Integrate in θ .**Step 11.1:**Use $\tan^2 \theta = \sec^2 \theta - 1$

$$\begin{aligned}\tan^2 \theta &= \sec^2 \theta - 1 \\ \int \tan^2 \theta \, d\theta &= \int (\sec^2 \theta - 1) \, d\theta \\ &= \int \sec^2 \theta \, d\theta - \int 1 \, d\theta\end{aligned}$$

Step 11.2:

Integrate

$$\begin{aligned}\int \sec^2 \theta \, d\theta &= \tan \theta \\ \int 1 \, d\theta &= \theta \\ \int \tan^2 \theta \, d\theta &= \tan \theta - \theta + C\end{aligned}$$

Step 12: Convert back to x .

Step 12.1:

Find θ

$$x = 2 \sin \theta$$

$$\frac{1}{2}x = \frac{1}{2}2 \sin \theta$$

$$\frac{x}{2} = \sin \theta$$

$$\theta = \sin^{-1}\left(\frac{x}{2}\right)$$

Step 12.2:

Find $\tan \theta$ using

[basic trig ratios](#)

$$\sin \theta = \frac{x}{2}$$

$$\text{opposite} = x, \quad \text{hypotenuse} = 2$$

$$\text{adjacent}^2 = 2^2 - x^2 = 4 - x^2$$

$$\text{adjacent} = \sqrt{4 - x^2}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{x}{\sqrt{4 - x^2}}$$

Step 13: Write the final answer and choose the correct option.

Step 13.1:

Substitute back

$$\begin{aligned} \int \frac{x^2}{(4 - x^2)^{\frac{3}{2}}} dx &= \tan \theta - \theta + C \\ &= \frac{x}{\sqrt{4 - x^2}} - \sin^{-1}\left(\frac{x}{2}\right) + C \end{aligned}$$

Step 13.2:

Final answer

The correct option is **b**.

Definitions and Notation

Antiderivative

Definition: An antiderivative of a function is another function whose derivative gives back the original function. In simple words: if you take the derivative of the new function, you return to the old one.

Example:

$$\frac{d}{dx} \left(\frac{x^3}{3} \right) = x^2,$$

so one antiderivative of x^2 is $\frac{x^3}{3}$.

Denominator

Definition: In a fraction $\frac{p(x)}{q(x)}$, the denominator is the bottom part, $q(x)$.

Example:

$$\frac{x^2}{(4 - x^2)^{3/2}}$$

has numerator x^2 and denominator $(4 - x^2)^{3/2}$.

Numerator

Definition: In a fraction $\frac{p(x)}{q(x)}$, the numerator is the top part, $p(x)$.

Example:

$$\frac{x^2}{(4 - x^2)^{3/2}}$$

has numerator x^2 .

Constant

Definition: A constant is a fixed, unchanging value or number, such as 5, -10 , or π , that remains the same throughout a specific expression, equation, or function.

Example: In

$$3x + 7,$$

the 7 is the constant term.

Variable

Definition: A variable is a symbol, typically written in a lowercase letter like x , y , or z , that represents an unknown numerical value or a quantity that can change.

Example: In

$$3x + 7,$$

the x is the variable.

Power of a Product Rule

Definition: When raising a product of terms to a power, apply that power to each factor inside the parentheses, expressed as

$$(ab)^n = a^n b^n.$$

Example:

$$(3x)^2 = 3^2 x^2 = 9x^2.$$

Trigonometric Substitution

Definition: Trigonometric substitution is a method for simplifying integrals with square roots by replacing x with a trig expression involving an angle.

Example:

$$\sqrt{4 - x^2} = \sqrt{2^2 - x^2}$$

matches the pattern $\sqrt{a^2 - x^2}$, so we use $x = 2 \sin \theta$.

Common patterns:

$$\sqrt{a^2 - x^2} : x = a \sin \theta, \quad \sqrt{a^2 + x^2} : x = a \tan \theta, \quad \sqrt{x^2 - a^2} : x = a \sec \theta.$$

Fractional Exponents

Definition: Fractional exponents rewrite powers as roots.

$$a^{m/n} = \sqrt[n]{a^m} = \left(a^{1/n}\right)^m.$$

Example:

$$(4 - x^2)^{3/2} = \left((4 - x^2)^{1/2}\right)^3 = \left(\sqrt{4 - x^2}\right)^3.$$

This means the $3/2$ power tells us to take a square root first, then cube.

Square of Trigonometric Functions

Definition:

$$(\sin \theta)^2 = \sin^2 \theta, \quad (\cos \theta)^2 = \cos^2 \theta, \quad (\tan \theta)^2 = \tan^2 \theta.$$

Pythagorean Identity

Definition: The Pythagorean identity is the trig fact

$$\sin^2 \theta + \cos^2 \theta = 1 \quad \Longleftrightarrow \quad 1 - \sin^2 \theta = \cos^2 \theta.$$

Example:

$$4 - 4 \sin^2 \theta = 4(1 - \sin^2 \theta) = 4 \cos^2 \theta.$$

Square Root

Definition:

$$y^2 = x \quad \Longrightarrow \quad y \text{ is a square root of } x.$$

For nonnegative real numbers,

$$\sqrt{x} = y \iff y^2 = x \text{ and } y \geq 0.$$

Example:

$$\sqrt{9} = 3$$

because

$$3^2 = 9.$$

Key Properties:

$$\sqrt{x^2} = |x|$$

$$\sqrt{a}\sqrt{b} = \sqrt{ab}$$

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

$$(\sqrt{a})^2 = a \quad \text{and} \quad a^{\frac{1}{2}} = \sqrt{a}$$

Constant Multiple Rule

Definition: When taking a derivative, a constant factor can stay in front:

$$\frac{d}{d\theta}(c f(\theta)) = c f'(\theta).$$

Example:

$$\frac{d}{d\theta}(2 \sin \theta) = 2 \cos \theta.$$

Absolute Value

Definition: The absolute value of a number is its distance from 0, so it is never negative.

Example:

$$|3| = 3, \quad |-3| = 3.$$

In Q34 we use

$$\sqrt{u^2} = |u|.$$

Since $0 \leq \theta \leq \frac{\pi}{2}$, we know $\cos \theta \geq 0$, so

$$|\cos \theta| = \cos \theta.$$

Basic Trig Ratios: Sine, Cosine, and Tangent

Definition: In a right triangle:

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \quad \tan \theta = \frac{\text{opposite}}{\text{adjacent}}.$$

Example: If $\sin \theta = \frac{x}{2}$, then opposite = x and hypotenuse = 2. The adjacent side is

$$\sqrt{2^2 - x^2} = \sqrt{4 - x^2},$$

so

$$\tan \theta = \frac{x}{\sqrt{4-x^2}}.$$

Inverse Sine $\sin^{-1}(x)$

Definition: Also written $\arcsin(x)$, inverse sine returns the angle whose sine is x .

Example: If $\sin \theta = \frac{x}{2}$, then

$$\theta = \sin^{-1}\left(\frac{x}{2}\right).$$